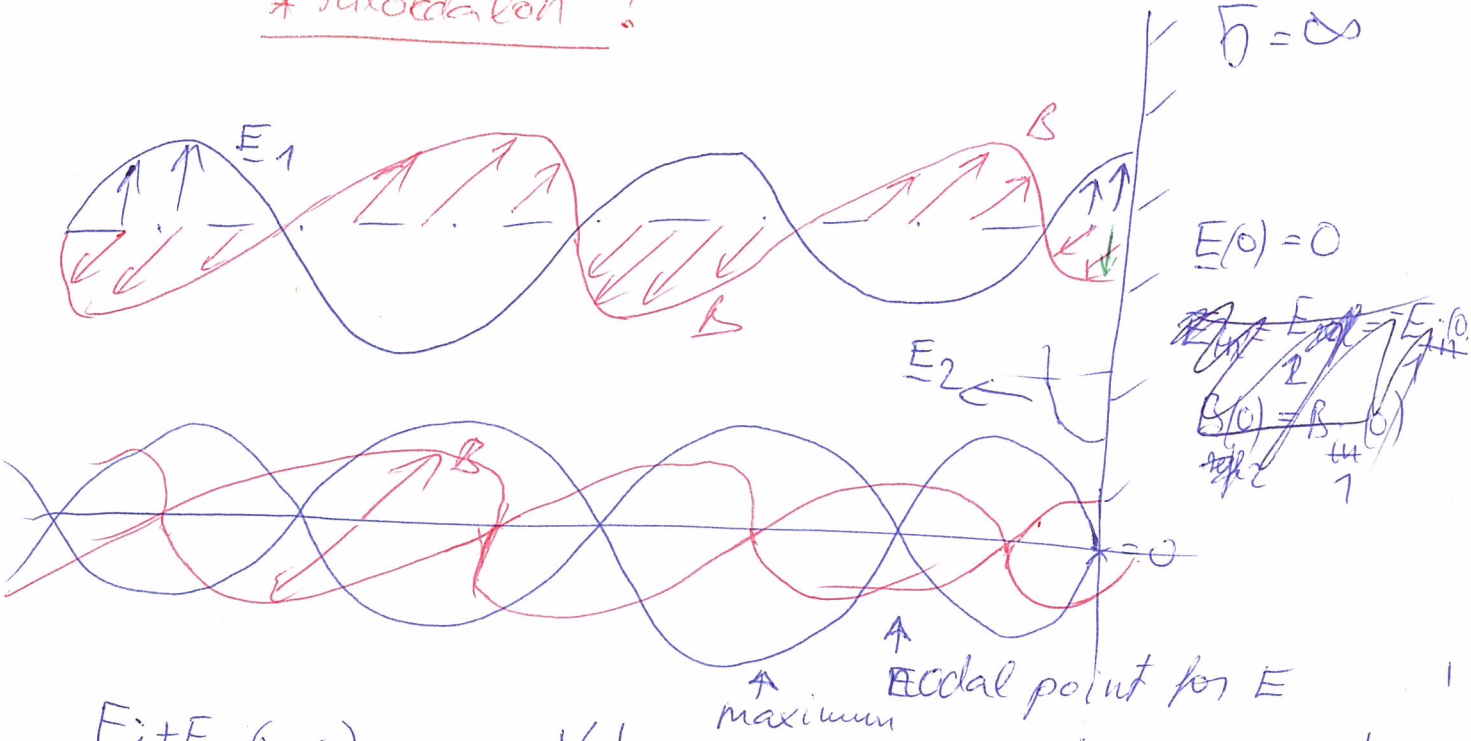
Standing EM. waves

What happens to an EM plane wave reflecting back on a perfectly conducting surface?

* filtration !



$E_{in} + E_R(x=0) = 0 \quad \forall t$ $\quad -$: π phase change!

$E_{in} = E_0 \cdot e^{i(kx - \omega t)}$

$$E_R = -E_0 \cdot e^{i(-kx - \omega t)}$$

$$E_{in} + E_R = E_0 \cdot e^{i(kx - \omega t)} - E_0 \cdot e^{i(-kx - \omega t)} =$$

$$= E_0 \cdot e^{-i\omega t} \cdot \left(e^{ikx + \pi} - e^{-ikx} \right) = E_0 \cdot 2i \sin(kx) \cdot e^{-i\omega t}$$

$$= 2E_0 \cdot \sin(kx) \cdot e^{i\left(\omega t + \frac{\pi}{2}\right)}$$

$$\underline{B} = \frac{\underline{k} \times \underline{E}}{\omega} \qquad Re\left(E_1 + E_2\right) = 2E_0 \sin kx \cdot \sin \omega t$$

→ Maxima of B coincide with nodal points of E and v.v.

↑ Standing wave

Where is energy ~~flowing~~ going?

Nowhere! Oscillating

* Reflection on a planar interface :

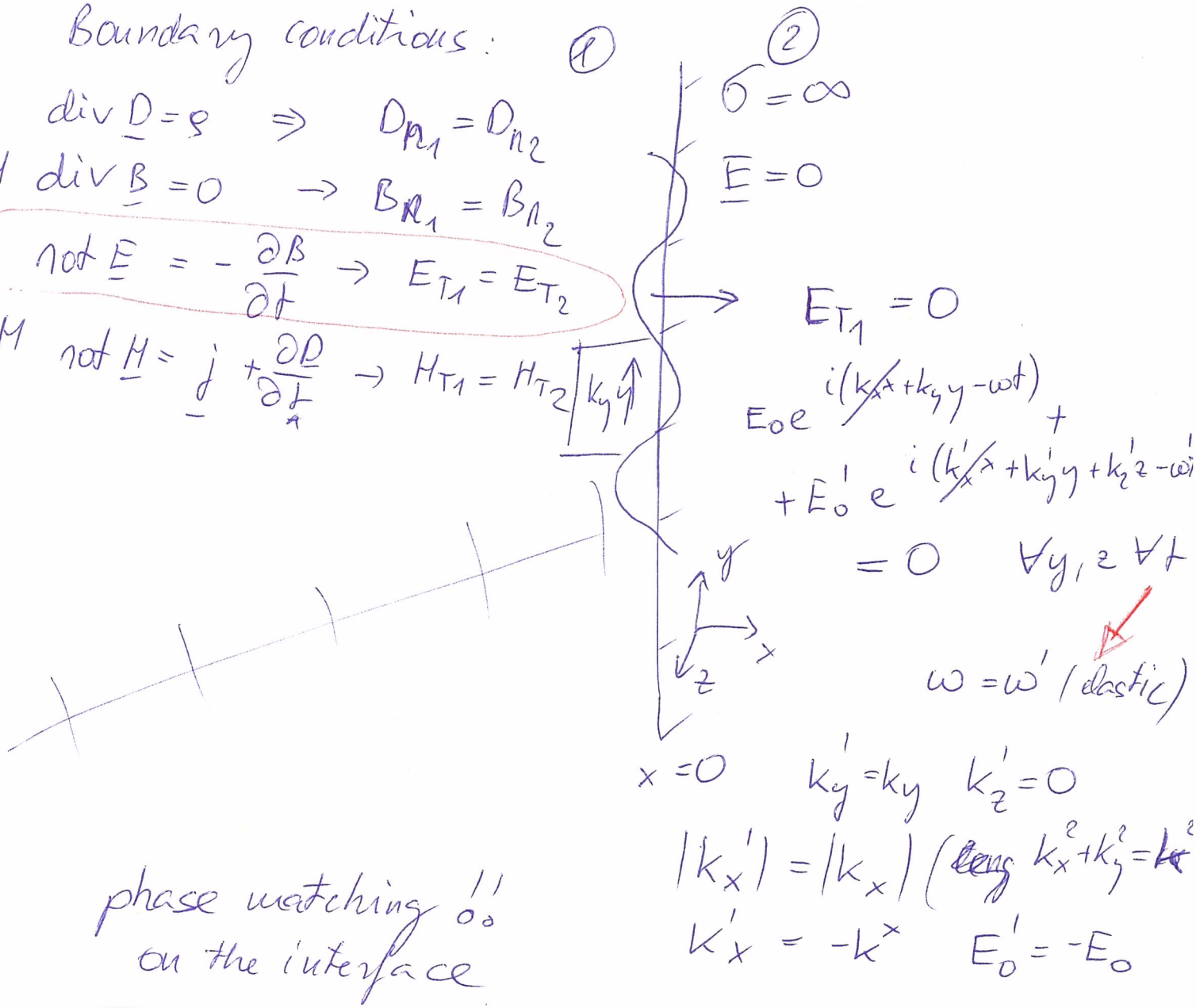
Boundary conditions :

G $\operatorname{div} \underline{D} = \rho \Rightarrow D_{n_1} = D_{n_2}$

GM $\operatorname{div} \underline{B} = 0 \rightarrow B_{n_1} = B_{n_2}$

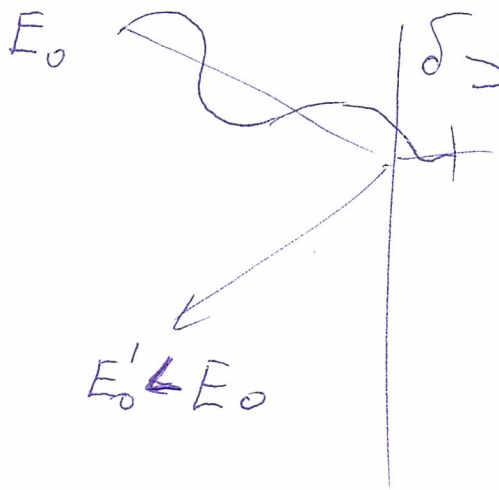
F $\operatorname{rot} \underline{E} = -\frac{\partial \underline{B}}{\partial t} \rightarrow E_{T_1} = E_{T_2}$

AM $\operatorname{rot} \underline{H} = \underline{j} + \frac{\partial \underline{D}}{\partial t} \rightarrow H_{T_1} = H_{T_2}$



phase matching !! on the interface

If σ = finite : skin effect

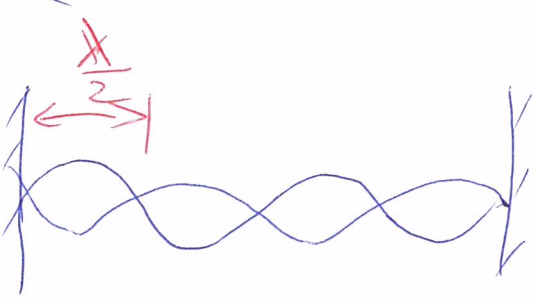


Boundary conditions are more complex, H is also involved...

16/2

Let's close the space with another conductor (mirror!)

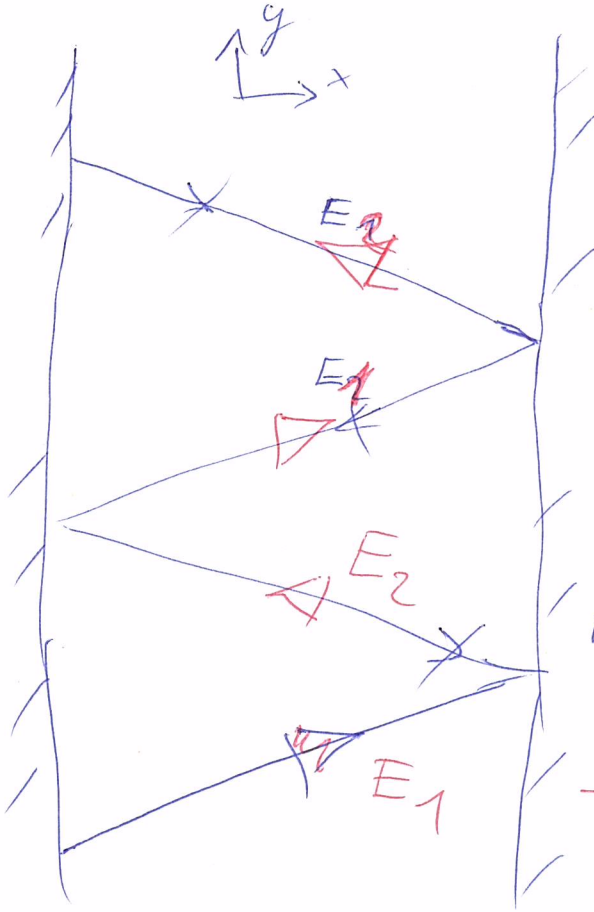
(not a perfect mirror: → skin depth)



Resonant cavity

EM standing wave can oscillate in it, if $L = n \cdot \frac{\lambda}{2} = \frac{n}{\cancel{4}} \cdot \cancel{2\pi} \frac{\lambda}{2\pi} \pi$

$$\boxed{k \cdot L = n \cdot \pi}$$



If non-orthogonal incidence:

$(k_x, k_y, 0)$

$E_1 = E_0 \cdot e^{i(k_x x + k_y y - \omega t)}$

$E_2 = -E_0 \, e^{i(-k_x x + k_y y - \omega t)}$

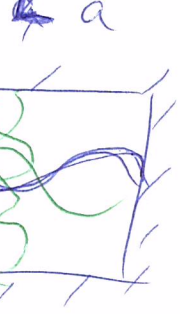
$E_1 + E_2 = 2E_0 \cdot i \sin(k_x x) \cdot e^{i(k_y y - \omega t)}$

$k_x \cdot L = n\pi$ standing wave in x; propagation in y

Waveguide! (microwaves)

Fiber optics: total internal reflection!

Let us close it in y



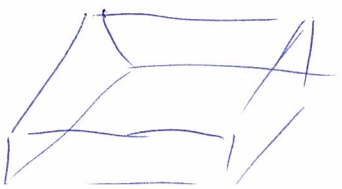
Standing in x and y

$k_x L = n_x \pi$

$k_y b = n_y \pi$

$e^{i(k_z z - \omega t)}$

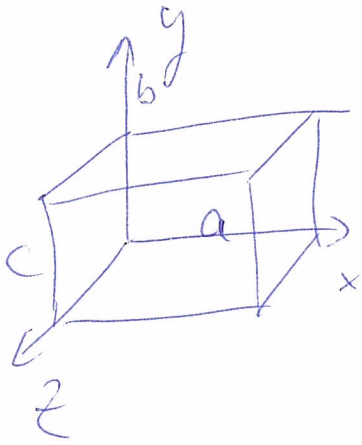
If we close it in z: resonator cavity



$k_z = \downarrow$

$+ k_z C = n_z \pi$ no propagation

Microwave oven!

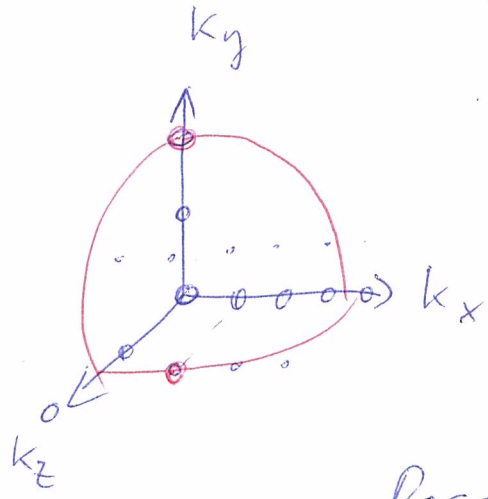


$$E = E_0 \cdot \sin(k_x \cdot a) \cdot \sin(k_y b) \cdot \sin(k_z c) \cdot e^{-i\omega t}$$

$$k_x a = n_x \pi \rightarrow \quad k_x = \frac{n_x \pi}{a}$$

$$k_y b = n_y \pi \qquad k_y = \frac{n_y \pi}{b}$$

$$k_z c = n_z \pi \qquad k_z = \frac{n_z \pi}{c}$$



Which are the real solutions?

$$k_x^2 + k_y^2 + k_z^2 = k^2 \left(= \frac{\omega^2}{c^2} \right)$$

those that lye on a sphere of radius k

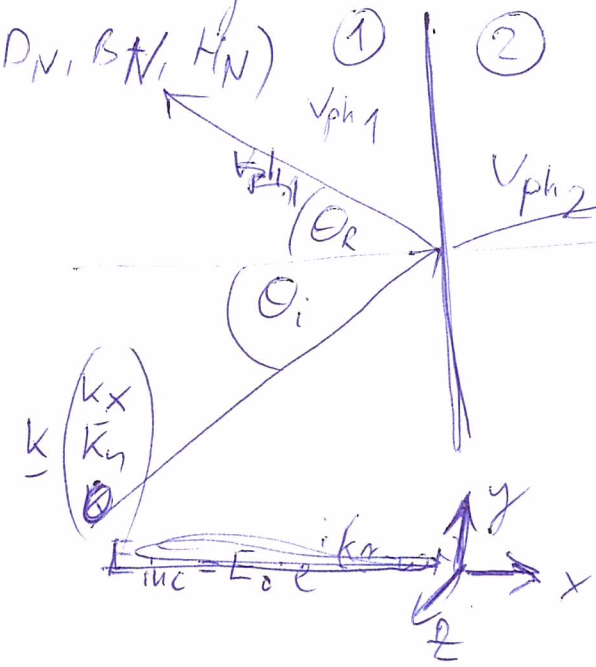
Resonant modes of a cavity !

Different for different frequencies.

Refraction at a planar interface in dielectrics

Same boundary conditions

(E_T, D_N, B_N, H_N)



$$v_{ph} = \frac{c}{\sqrt{\mu_{rel}\varepsilon_{rel}}} = \frac{c}{n}$$

$$E_{inc} = E_0 \cdot e^{i(\underline{k}\underline{r} - \omega t)}$$

$$E_{refl} = E_0' \, e^{i(\underline{k}'\underline{r} - \omega t)}$$

$(\omega = \omega' = \omega''$ obvious$)$

$k_z' - k_z'' = k_z = 0$ occ. together.
(no oscillation in z)

$$E_{inc}(x=0) + E_{refl}(x=0) = E_{trans}(x=$$

$$E_0 e^{i(k_x x + k_y y - \omega t)} + E_0' e^{i(k_x' x + k_y' y - \omega t)} = E'' e^{i(k_x'' x + k_y'' y - \omega$$

$$k_y = k_y' = k_y''$$

The lenght of k changes!

$\theta_i = \theta_r$

suppose $V_2 < V_1$

$$k_y = k_y''$$

$$k \sin\theta_i = k'' \cdot \sin\theta_{trans}$$

$$\frac{c}{n_1} = V_1 = \frac{\omega}{k} \qquad V_2 = \frac{\omega}{k''} = \frac{c}{n_2}$$



$$k = \frac{\omega}{c} \cdot n_1 \qquad k'' = \frac{\omega}{c} n_2$$

$k'' > k$

$$\frac{k}{k''} = \frac{\sin\theta_{trans}}{\sin\theta_{in}}$$

$$\frac{\frac{\omega}{c} n_1}{\frac{\omega}{c} n_2} = \frac{\sin\theta_t}{\sin\theta_{in}}$$

Law of refraction (Snell's – Descartes)

The amplitude of the reflected and refracted wave depends on the polarization, the inc angle and the materials (n_1, n_2)

<u>In general all S waves are present</u>

Interesting special cases:

transmitted wave is absent: Total internal reflection

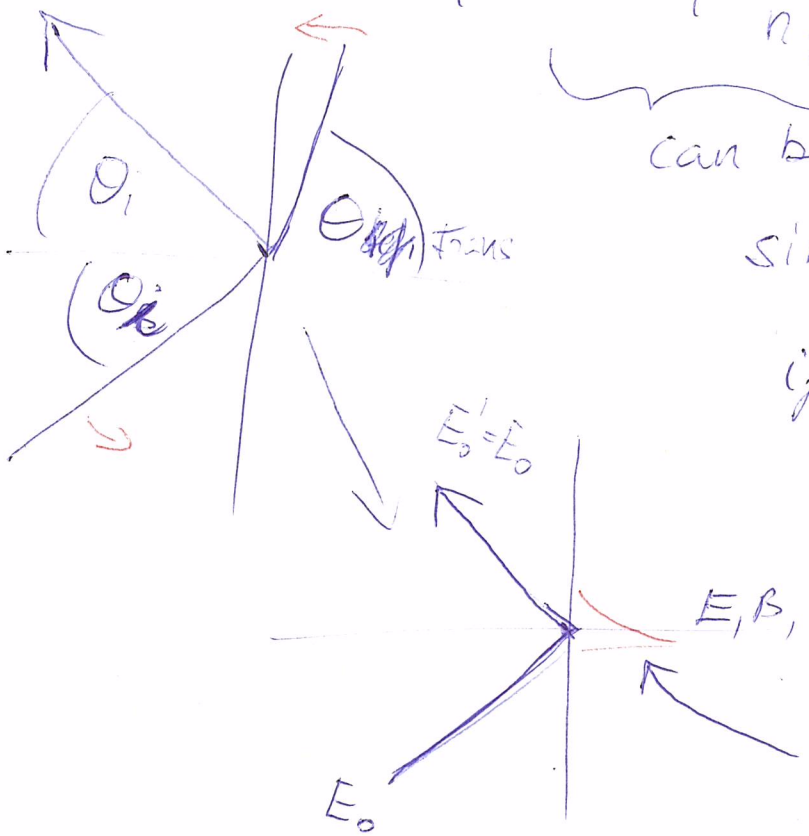
$$\sin\Theta_t = \sin\Theta_i \frac{n_1}{n_2} \qquad n_1 > n_2$$

$\sin\Theta_i \frac{n_1}{n_2}$ can be larger than 1

$$\sin\Theta_i \frac{n_1}{n_2} > 1$$

if $\sin\Theta_i > \frac{n_2}{n_1}$ possible

↑ no solution for Θ_t



$E_1 B_1 = 0$ does not fulfill the bound. conditions!

evanescent wave

$\cos\Theta_t$ decribes $k_x' = k' \cos\Theta_t$

$$\cos\Theta_t = \sqrt{1 - \sin^2\Theta_t} = \sqrt{1 - \sin^2\Theta_{in} \frac{n_1}{n_2}} =$$

$$= i\sqrt{\sin^2\Theta_i \frac{n_1}{n_2} - 1} = i\gamma$$

$\sin^2\Theta_i \frac{n_1}{n_2} > 1$

negative

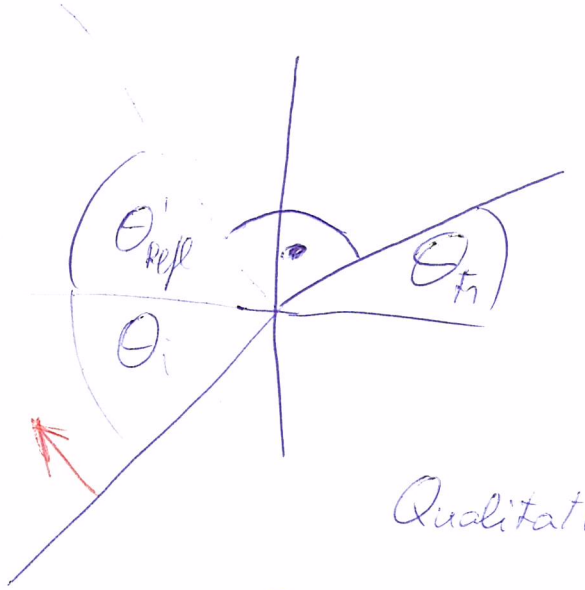
$$E_t = E_0'' \cdot \underbrace{e^{i(jk''x + k_y'' y - \omega t)}}_{e^{-\gamma k'' x}} \cdot \underbrace{e^{i(k_y'' y - \omega t)}}_{\text{propagation in } y!}$$

$e^{-\gamma k'' x}$: exp decay

Just like the ideal conductor, except:

- the evan. wave propagates in the low ind. mat.
- there is a phase jump at the interface

Other special case : Brewster effect



If the wave is polarized in the plane of inc. (P polarization or TM) (as opposed to S or TE

Qualitative explanation:

$\underline{E}$ → oscillating dipole // $\underline{E}$ $\underline{dc}$ // $\underline{E}$

↳ reflection can be modelled as the radiation of billions of oscillating dipoles.

But: radiation of the dipol towards $\underline{dc}$ is zero!

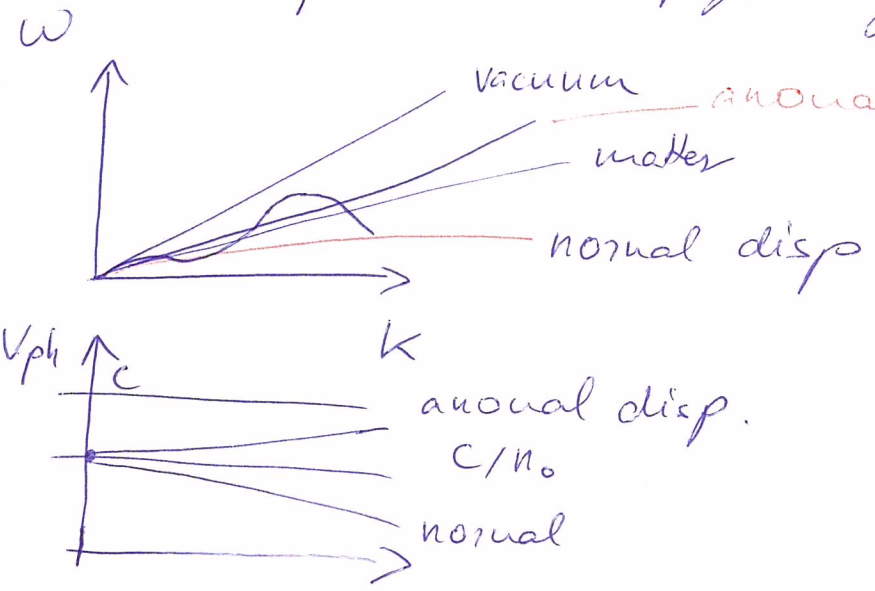
More exact to say that the solution of the boundary value problem gives $\emptyset$

Dispersion of EM waves

$$v_{ph} = \frac{\omega}{k} = \frac{1}{\sqrt{\varepsilon\mu}} = \frac{1}{\sqrt{\varepsilon_0\mu_0}} \cdot \frac{1}{\sqrt{\varepsilon_{rel}\mu_{rel}}} = \frac{c}{n(\omega)}$$

($\sqrt{\varepsilon_0\mu_0} = c$, $\frac{1}{\sqrt{\varepsilon_{rel}\mu_{rel}}} = \frac{1}{n}$)

ε: polarizibility of bound electrons!
depends on the frequency!



$\omega(k)$ dispersion relation

Effect on wave:

① Groupe velocity! ② Refraction, prism, rainbow...

Let's investigate the combination of 2 plane waves.

$$E = E_1 + E_2 = E_0 e^{i(k_1 x - \omega_1 t)} + E_0 e^{i(k_2 x - \omega_2 t)} =$$

$$= E_0 e^{i\frac{(k_1 x + k_2 x - \omega_1 t - \omega_2 t)}{2}} \left[e^{i(k_1 x - k_2 x - \omega_1 t + \omega_2 t)} + \right.$$

$$\left. + e^{-i(k_1 x - k_2 x - \omega_1 t + \omega_2 t)} \right] =$$

(4)

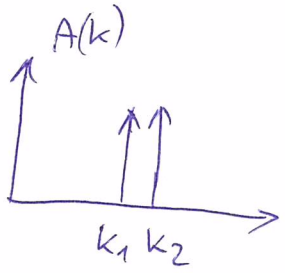
$$E = E_0 \cdot e^{i\left(\frac{k_1+k_2}{2}x - \frac{\omega_1+\omega_2}{2}t\right)} \cdot 2\cos\frac{k_1-k_2}{2}x - \frac{(\omega_1-\omega_2)}{2}t$$

↑ average wave number; ↑ average ω

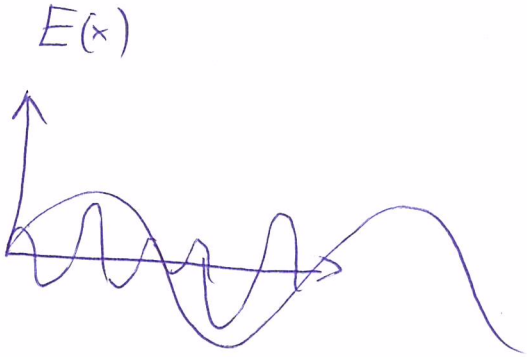
envelope propagating at $V_{env} = \frac{\omega_1-\omega_2}{k_1-k_2}$

$$V_g = \frac{\Delta\omega}{\Delta k} = \frac{\partial\omega}{\partial k}$$

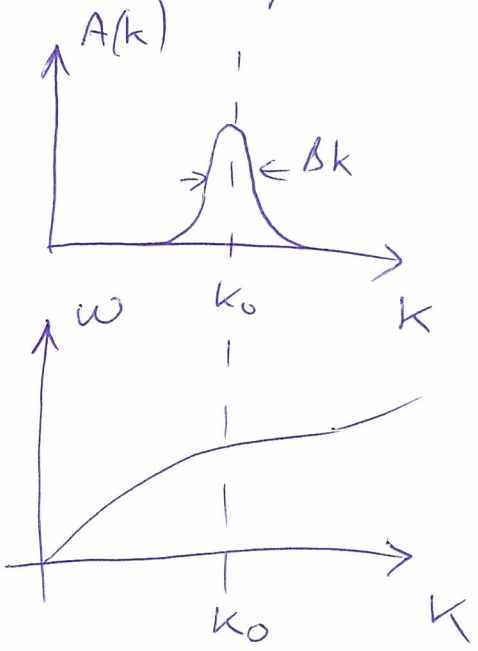
For these 2 waves:



IFT →



For a wave packet:



Taylor series around k_0

$$\omega(k) = \omega(k_0) + \frac{\partial\omega}{\partial k}\Big|_{k_0}(k-k_0) + \frac{\partial^2\omega}{\partial k^2}\cdots$$

↑ group velocity; ↑ envelope distortion

Challenge 19?